CIE AS PHYSICS (9702) • QUESTION PAPER

7.5 Polarisation

Syllabus Section: 7. Waves **Paper Type:** Multiple Choice (Paper 1) **Total:** 14 Questions (14 Marks)

Question 1 Easy [1 mark]

The following are statements about waves. Which statement below describes a situation in which polarisation would not happen?

A microwaves pass through a metal grid

B sound waves pass through a metal grid

C light waves are reflected

D light waves are scattered

Question 2 Easy [1 mark]

A student looked at the liquid crystal display on their calculator with a polarising film. They observed that as they rotated the film, the display changed.

Which property of the radiation from the calculator display is described correctly?

A the emitted radiation is a transverse wave

B the emitted radiation is a wave with 3 cm wavelength

C the emitted radiation is unpolarised

D the emitted radiation is a longitudinal wave

Question 3 Easy [1 mark]

Which of the following waves can be polarised?

A transverse

B progressive

C stationary

D longitudinal

Question 4 Easy [1 mark]

Malus's law shows the intensity of polarised light as it emerges through an analyser. Which is the correct equation to calculate the intensity?

A $I_0 \sin^2 \theta$

B $I_0 \cos^2 \theta$

C $I_0 \cos \theta$

D $\frac{I_0}{2} \cos^2 \theta$

Question 5 Easy [1 mark]

A polarised light beam emerges from a polariser. It is then transmitted onto an analyser, which is at angle ? to the polariser.

What does the value of the angle ? need to be, so that the light emerging from the analyser is at maximum intensity?

A 45°

B 90°

C 180°

D 270°

Question 6 Medium [1 mark]

A linearly polarised electromagnetic beam has an intensity of 16 W m^{-2} with an angle of incidence of 45° to the vertical.

What is the intensity of the transmitted beam?

A 20 W m^{-2}

B 12 W m^{-2}

C 16 W m^{-2}

D 8 W m^{-2}

Question 7 Medium [1 mark]

Unpolarised light of intensity I_0 is incident on a polariser with a vertical transmission axis. The transmitted light is incident on a sheet of material X. After transmission through X the intensity of light is $\frac{I_0}{2}$.

It is suggested that X could be

1	a polariser with a vertical transmission axis
2	a polariser with a horizontal transmission axis
3	non-polarising glass

A 1 & 2

B 2 & 3

C 1 & 3

D 3 only

Question 8 Medium [1 mark]

The quality of the sound produced by a portable radio can be improved by changing the orientation of the aerial.

Which statement will correctly describe the explanation of this improvement?

A the radio waves become polarised as a result of adjusting the aerial

B the radio waves become unpolarised as a result of adjusting the aerial

C the radio waves from the transmitter are unpolarised

D the radio waves from the transmitter are polarised

Question 9 Medium [1 mark]

Unpolarised light of intensity I is incident upon a polariser. The light is then transmitted to a second polariser and then a third. The polarisers have the same orientation.

Which statement is true?

A The intensity of the light emerging from the second polariser is $\frac{1}{4}I$

B The intensity of the light emerging from the second polariser is $2I$

C The intensity of the light emerging from the third polariser is $\frac{1}{8}I$

D The intensity of the light emerging from the third polariser is $\frac{1}{2}I$

Question 10 Medium [1 mark]

Unpolarised light of intensity 8 W m^{-2} is incident upon a polariser. The light is then transmitted to a second polariser at an angle of 25° to that of the first polariser.

What is the intensity of the light beam emerging from the second?

A 1.6 W m^{-2}

B 3.3 W m^{-2}

C 3.6 W m^{-2}

D 6.6 W m^{-2}

Question 11 Hard [1 mark]

A beam of unpolarised light is incident on the first of two parallel polarisers. The transmission axis of the two polarisers are initially parallel.

The first polariser is now rotated about the direction of the incident beam by an angle smaller than 90°.

Which row in the table describes the changes to the intensity and polarisation of the light transmitted through both polarisers?

	Intensity	Polarisation
A	no change	no change
B	no change	different
C	different	different
D	different	no change

A

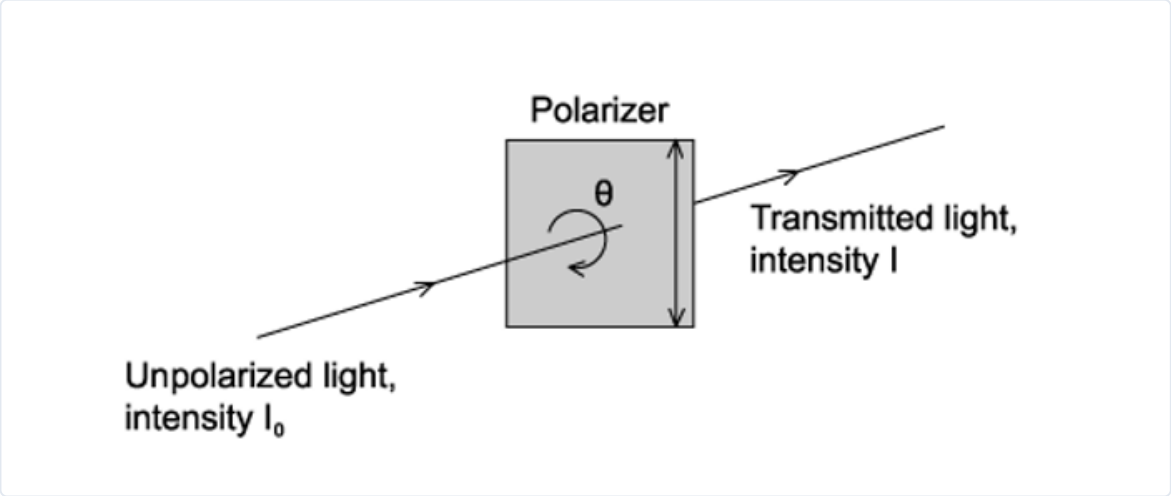
B

C

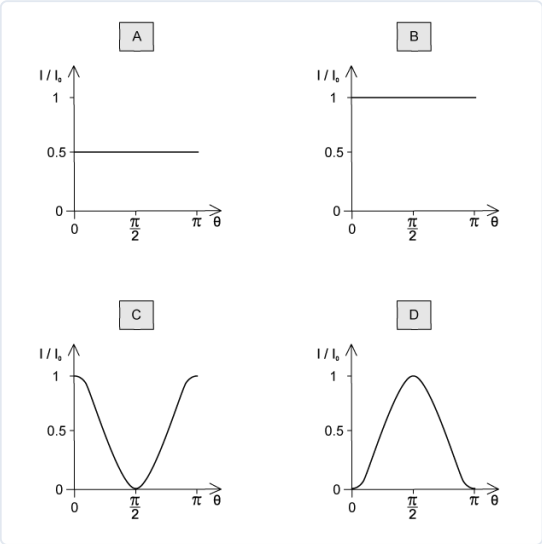
D

Question 12 Hard [1 mark]

Unpolarised light of intensity I_0 is incident on a polariser that has a vertical transmission axis, as shown in the diagram below.



Which graph shows the intensity of the transmitted light ?



A

B

C

D

Question 13 Hard [1 mark]

Unpolarised light is incident on a polariser. The light transmitted by the first polariser is then incident on a second polariser. The polarising axis of the second sheet is 60˚ to that of the first.

The intensity emerging from the second polariser is I_2.

Which of the following gives the intensity incident on the first polariser?

A $\frac{I_2}{4}$

B $\frac{I_2}{8}$

C $8I_2$

D $4I_2$

Question 14 Hard [1 mark]

Unpolarised light of intensity I_0 is incident on the first of two polarising sheets. Initially the planes of polarisation of the sheets are perpendicular.

Which sheet must be rotated and by what angle θ, so that light intensity $\frac{I_0}{4}$ can emerge from the second sheet ?

	Rotated sheet	Angle θ of rotation
A	1 or 2	$\cos^{-1}\frac{1}{2}$
B	1 or 2	$\cos^{-1}\frac{\sqrt{2}}{2}$
C	2 only	$\cos^{-1}\ \frac{1}{2}$
D	1 only	$\cos^{-1}\frac{\sqrt{2}}{2}$

A

B

C

D